

BeeSMART Manual

Version 3.0



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1 Introduction

BeeSMART is a Wi2009Fi based honey dispensing system focused on few components, simple operation and relatively low cost. The system can of course also be used for other liquids where weight-based filling is desired.

The system is browser-based and requires no app. Access it from a PC, tablet or smartphone as long as the device can connect to the BeeSMART Wi2009Fi.



The system typically consists of:

- BeeSMART module with integrated Wi2009Fi control and servo
- Servo horn and pushrod
- BeeSMART scale (5 kg)
- USB-C power supply
- Mounting bracket and inserts for different taps

The mounting bracket fits taps with a collar of about 54 mm diameter and at least 10 mm width. Inserts for 40 mm collars are included.

2 Parts List



Contents of a standard BeeSMART kit:

2 ×	M3 self-locking nut
2 ×	M4 nut
1 ×	M3 × 16 screw
1 ×	M3 × 25 screw
2 ×	M4 × 25 screw
1 ×	BeeSMART module with servo and Wi2009Fi control
1 ×	Pushrod
1 ×	Servo horn, extension and screw
2 ×	Inserts for 40 mm tap
1 ×	5 kg BeeSMART scale
1 ×	USB-C power supply

3 Video material

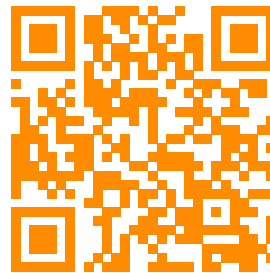
There are videos for BeeSMART covering:

- Assembly and mounting of BeeSMART
- Demonstration of dispensing functions
- Usage examples with different settings

The videos are especially useful during initial setup and for fine-tuning the servo and control parameters.



BeeSMART in use



BeeSMART in use



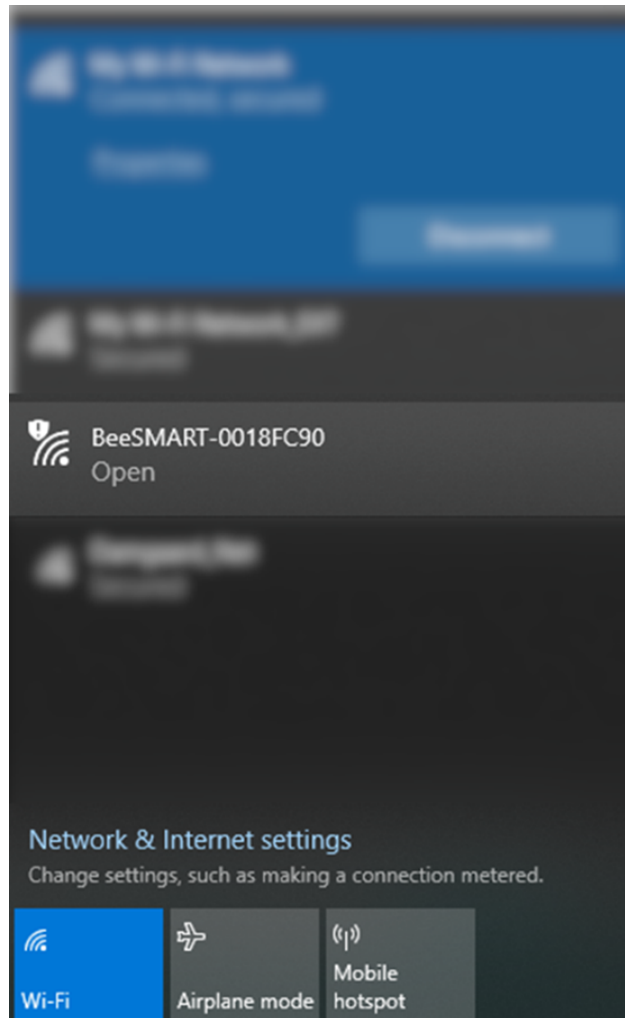
Advanced features in 3.0



Tap and bracket

4 User guide

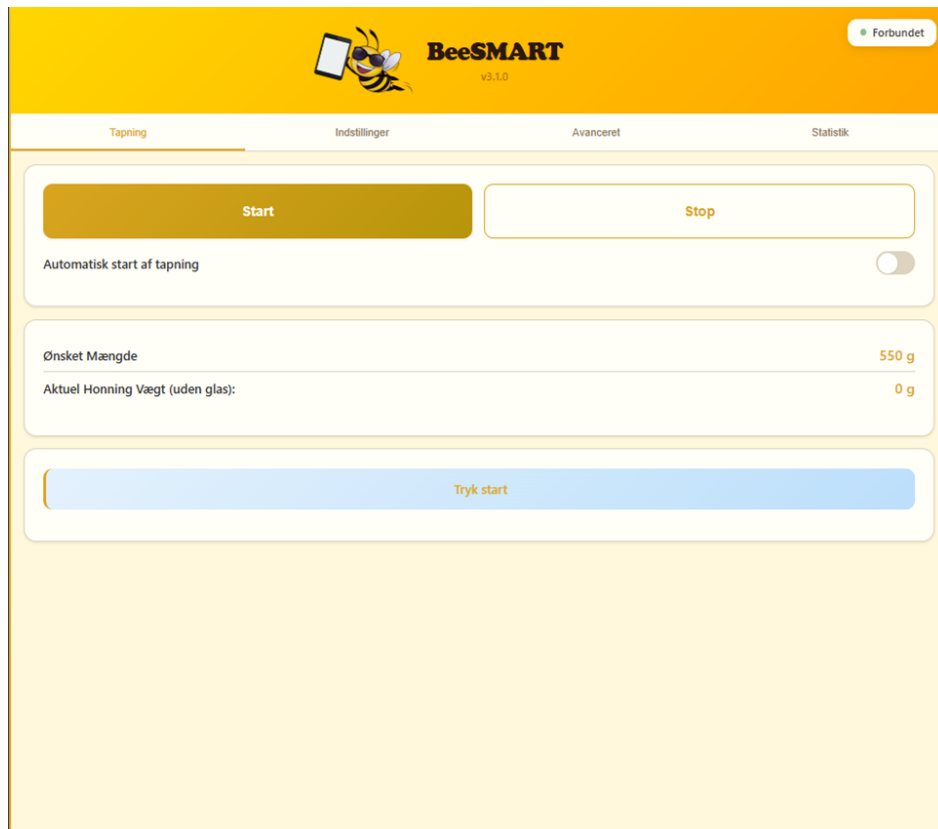
After connecting USB power, the servo and the scale, BeeSMART will create a Wi2009Fi Access Point. This should appear within approximately 30 seconds. If not, briefly press the reset button on the module.



Connect to the displayed BeeSMART network from a PC, tablet or smartphone. In many cases a browser will open automatically with the BeeSMART interface. If this does not happen, open a browser manually and enter the following address:

`beesmart.local`

Alternatively, the IP address 192.168.4.1 can also be used.



The main screen provides an overview of:

- Desired quantity to be dispensed (excluding jar)
- Current weight on the scale
- Status text at the bottom describing what the system is doing

The **Start** button begins a dispense cycle, while **Stop** interrupts a running dispense.

The **Automatic start of dispensing** feature can be enabled via a switch. When enabled, BeeSMART will automatically start a new dispense when a new empty jar is detected on the scale.

BeeSMART v3.1.0

Forbundet

Tapping Indstillinger Avanceret Statistik

Ønsket Mængde

50g 550g 1000g

550

Luk hanen når der mangler

2

Glasregistrering: Glas vejer mere end

10

Max tappemængde

1000

Tare kun med tom vægt !

Aktuel Vægt: 0 g

Tare

Sprog

Dansk Deutsch English

On the *Settings* page you can configure:

- **Desired quantity** (honey without jar)
- **Close tap when X g remain**
- **Jar detection: Jar weighs more than**
- **Max dispense quantity**
- **Tare** (reset scale)
- **Language selection** (e.g. Danish, German, English)

Close tap when X g remain

This setting specifies how many grams before the desired quantity the tap is fully closed. It compensates for dripping and the last honey remaining after closing.

Jar detection

The jar weight threshold sets how heavy an empty jar must be to be detected. This prevents dispensing without a jar on the scale.

Max dispense quantity

This value increases the maximum of the *Desired quantity* slider if larger fills are needed. The default maximum the scale can handle is 5 kg.

Tare

Tare resets the scale. Use with an empty scale.

5 Advanced

On the *Advanced* page you can adjust control parameters (PID), the servo travel and the scale calibration.

The screenshot shows the BeeSMART v3.1.0 interface. The top navigation bar includes 'Tæpning', 'Indstillinger', 'Avanceret', and 'Statistik'. The 'Avanceret' section is active, displaying PID control settings, servo position limits, and weight calibration options. The 'PID Kontrol Parametre' section shows a viscosity slider set to 'Medium' and PID values Kp=8, Ti=7, Kd=2. The 'Indstil min og max Servo position' section shows servo range from 5° to 115°. The 'Vægt Kalibrering' section shows the current weight is 0g and the calibration weight is set to 283g.

The scale is calibrated at the factory. Calibration is normally not required unless significant deviations are observed.

BeeSMART uses PID-like control parameters to control servo opening relative to measured weight. Predefined viscosity profiles are available:

- **Low** – for thinner honey/liquid
- **Medium** – default
- **High** – for very thick honey
- **Custom** – free adjustment of Kp, Ti and Kd

Typical default values that often work well are:

- $K_p = 8$
- $T_i = 7$
- $K_d = 2$

Kp (Proportional)

K_p is multiplied by the difference between desired weight and current weight in the jar — i.e. the missing amount. A small K_p yields a small tap opening, while a larger K_p yields a larger opening for the same error.

- Larger K_p typically means faster filling.
- Too large K_p increases the risk of overfilling.

Ti (Integral/time constant)

T_i describes the system's *patience*. A small T_i means the system reacts quickly to missing honey and opens more. A large T_i causes slower reactions.

- Smaller T_i leads to faster correction.
- Too small T_i can cause overshoot and instability.

Kd (Derivative)

K_d reacts to how fast the weight is changing. If the jar fills very fast, K_d helps temporarily reduce servo opening so the rate slows. When the weight change becomes smaller again, K_d loses effect and the tap can open more.

- Too high K_d can cause an *open–close–open* behavior.
- K_d can help avoid overshoot when the source is nearly empty or the honey is relatively thin.
- Generally K_d can often be set to 0 unless extra damping is needed.

General recommendations

- Larger K_p = faster filling, but higher risk of overshoot.
- Smaller T_i = faster filling, but higher risk of overshoot.
- Increase K_d only cautiously and in small steps.

On the same page you can set *Servo Minimum* and *Servo Maximum*. Buttons are provided to move the servo to min and max positions for testing.

Setting servo travel

1. Remove the servo horn from the servo.
2. Ensure the tap is physically closed.
3. Set *Servo Minimum* to a suitable value and move the servo to the min position.
4. Mount the servo horn so the tap is just closed in the min position and tighten the screw.
5. Increase *Servo Maximum* gradually while testing the opening until a suitable full opening is reached.



The scale is delivered calibrated. Calibration is normally only necessary if systematic deviations are repeatedly observed.

On the page select a calibration weight (e.g. 50 g, 283 g or 1000 g) and follow the calibration procedure. NOTE: The scale must be empty when pressing "Calibrate". Use a as-accurate reference weight as possible.

6 Control parameters — Details

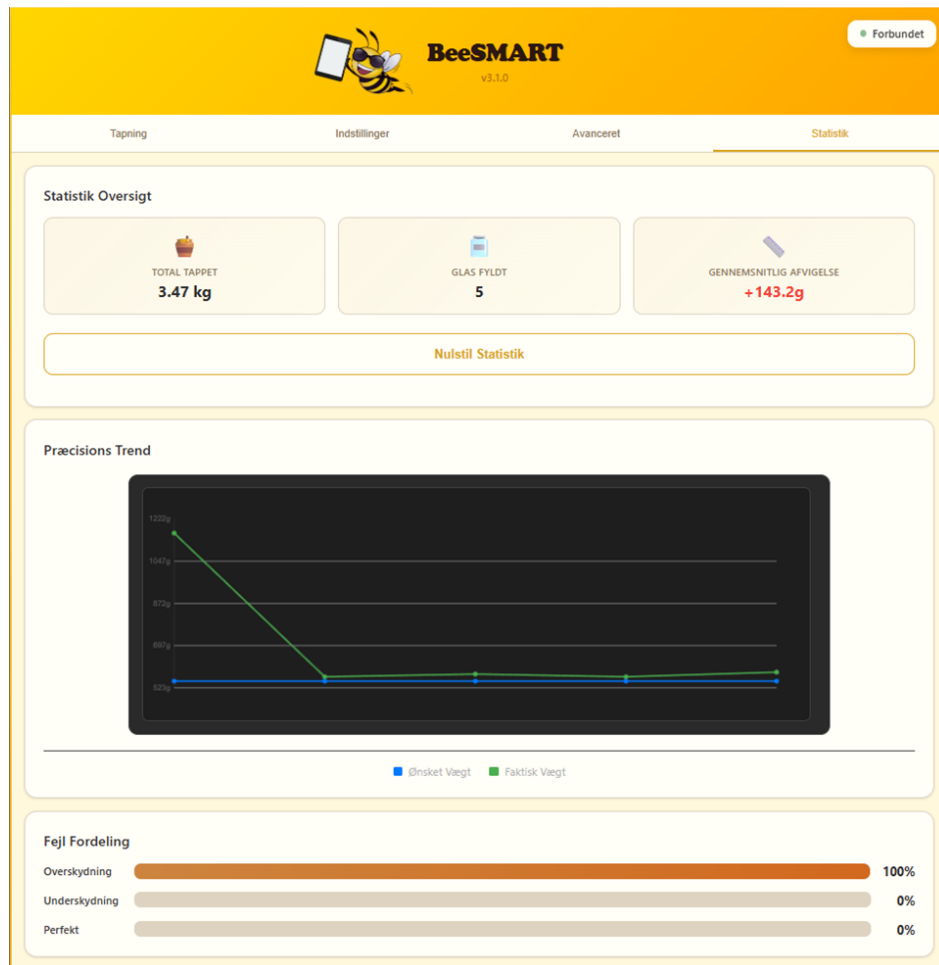
The control parameters K_p , T_i and K_d work together to control fill speed and ensure the desired weight is reached as accurately as possible.

- Always start from the default values.
- Adjust only one parameter at a time.
- Use the Statistics page to evaluate results over multiple fills.

- If fills typically end below the desired weight, increase K_p or T_i .
- If often overfilled, decrease K_p and/or increase T_i ; consider slightly increasing K_d .
- Very thick honey may require larger K_p to maintain good flow.

7 Statistics

The Statistics page provides an overview of how fills performed relative to the desired weight.



Typically shown:

- **Total dispensed volume** (e.g. in kg)
- **Number of jars filled**
- **Average deviation** in grams
- **Graph of last 10 fills** (desired vs actual)
- **Error distribution** — share of overfills, underfills and *perfect* fills

These data can be used to:

- Evaluate if control parameters are suitable
- Identify systematic overfilling or underfilling
- Support production documentation and quantity tracking